ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

NATURAL SCIENCES

PRACTICE EXAM 3 · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 8 Natural Sciences practice exam 3.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words if the science is correct. Accept other correct examples. Penalise missing units once per question.

SECTION A

QUESTION 1

1 1.1 B ✓ 1.2 C ✓ 1.3 A ✓ 1.4 B ✓ 1.5 D ✓ 1.6 A ✓ 1.7 C ✓ 1.8 C ✓ 1.9 A ✓ 1.10 D ✓ (10)

QUESTION 2

2.1 False ✓ – leaves ✓ (2)

2.2 False ✓ – Bacteria ✓ (2)

2.3 True ✓✓ (2)

2.4 True ✓✓ (2)

2.5 False ✓ – star ✓ (2)

TOTAL SECTION A: [20]

SECTION B

QUESTION 3

3.1 Any TWO: broad (large surface area) to absorb more light ✓; thin, so gases move in and out quickly ✓; many stomata for gas exchange; veins carry water to the cells (2)

3.2 Only the green parts contain chlorophyll ✓; chlorophyll is needed for photosynthesis, so starch was made only in the green parts ✓. (2)

3.3.1 Algae ✓ (1)

3.3.2 Small fish or ducks ✓ (1)

3.3.3 A pyramid with four levels ✓, algae at the wide bottom and kingfisher at the narrow top ✓, all levels labelled in the correct order ✓ (3)

3.3.4 Any ONE with explanation, e.g. fertiliser running into the pond causes too much algae to grow, which uses up oxygen so fish die ✓✓; pollution with chemicals or rubbish poisons the animals; draining the pond destroys the habitat (2)

3.4 Any TWO: thick stem that stores water ✓; spines instead of leaves to reduce water loss ✓; thick, waxy skin; long or widespread roots to collect water quickly (2)

3.5 Milk is heated (to kill harmful bacteria) and cooled ✓. Useful bacteria are added and the milk is kept warm ✓. The bacteria change the milk sugar into acid, which thickens the milk and gives it a sour taste ✓. (3)

3.6 Soap removes dirt and germs (micro-organisms) from your hands ✓, so they cannot enter your body through your mouth, nose or food ✓. (2)

3.7 Any TWO: viruses are much smaller ✓; viruses can only reproduce inside living cells ✓; antibiotics kill bacteria but not viruses; bacteria are cells, viruses are not; some bacteria are useful (2)

[20]

QUESTION 4

4.1.1 0 °C ✓ (1)

4.1.2 The heat energy is used to break the forces between the particles (to change the ice into water) ✓, not to make the particles move faster, so the temperature does not rise ✓. (2)

4.1.3 100 °C ✓ (1)

4.1.4 Liquid ✓ (1)

4.2 On hot days the concrete expands ✓; the gaps give it space to expand so that it does not crack or buckle ✓. (2)

4.3 (a) Mixture ✓ (b) Element ✓ (c) Compound ✓ (d) Mixture ✓ (4)

4.4 (a) 3 ✓ (b) 6 + 12 + 6 = 24 ✓ (2)

4.5 Metals: sodium and magnesium ✓ Non-metals: sulfur and oxygen ✓ (2)

4.6.1 Carbon dioxide ✓ (1)

4.6.2 Bubble the gas through clear limewater ✓; the limewater turns milky ✓. (2)

4.6.3 The carbon dioxide gas escapes into the air from the open beaker ✓ (1)

4.7 Physical change ✓ (1)

[20]

QUESTION 5

5.1 Electrons move from the wool to the ruler during rubbing ✓, so the ruler becomes negatively charged ✓. The charged ruler attracts the uncharged pieces of paper (it pulls on the charges in the paper) ✓. (3)

5.2 The bulb becomes brighter ✓ because the two cells give a bigger voltage, so a larger current flows ✓. (2)

5.3.1 Ammeter ✓ – measures current ✓ (2)

5.3.2 Voltmeter ✓ – measures the voltage (potential difference) across the bulb ✓ (2)

5.4 Any TWO: the wires and plugs overheat ✓; it can cause a fire ✓; it can trip the circuit breaker; appliances can be damaged (2)

5.5 $\text{(a) Angle of incidence} = 90° - 40° = 50°$ ✓; angle of reflection $= 50°$ ✓
$\text{(b) It reflects light from the Sun}$ ✓ (3)

5.6 Refraction ✓: light from the bottom bends away from the normal as it leaves the water, so the bottom seems higher (closer) than it really is ✓. (2)

5.7 Violet ✓ (1)

5.8 Red ✓, green ✓ and blue ✓ (3)

[20]

QUESTION 6

6.1 The distances between the planets are enormous compared with the sizes of the planets ✓; if the planets are drawn big enough to see, the distances will not fit on the paper ✓. (2)

6.2 Jupiter, Saturn, Uranus and Neptune ✓✓ (one mark for any two) (2)

6.3 Any TWO: liquid water ✓; the right distance from the Sun for suitable temperatures ✓; an atmosphere with oxygen; the ozone layer and magnetic field protect it from harmful radiation (2)

6.4 Starlight passes through moving layers of air in the atmosphere ✓, which bend (refract) the light by changing amounts, so the star seems to flicker ✓. (2)

6.5 Nebula → main-sequence star → red giant → white dwarf ✓✓✓ (3)

6.6 The light left Andromeda about 2,5 million years ago ✓, so we see the galaxy as it was 2,5 million years ago, not as it is now ✓. (2)

6.7 Any TWO: clear, dry skies in places such as the Karoo ✓; areas with very little light pollution ✓; a good view of the centre of the Milky Way and the southern sky; little radio interference in remote areas (2)

6.8 Any TWO: there is no air to breathe ✓; extreme heat and cold ✓; harmful radiation from the Sun; no air pressure; protection from tiny space particles (2)

6.9 A meteor is the streak of light seen when a piece of space rock burns up in the atmosphere ✓; a meteorite is a piece of space rock that reaches the Earth's surface ✓. (2)

6.10 The Vredefort Dome (Vredefort crater) ✓ (1)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

END OF MARKING GUIDELINE